MSc

Course title: Bayes Methods
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1. An 18×24 square-foot area of grass in a field was split into 3×3 square-foot cells (so $n = 48$). An indicator for army-worm damage was recorded for the centre square in each cell. When the grass was planted the plough and harrows ran down the columns.

	Easting							
Northing	1	2	3	4	5	6	7	8
1	1	1	1	0	0	0	0	1
2	1	1	1	0	0	1	0	1
3	0	1	1	1	1	0	1	1
4	1	1	1	0	0	1	1	1
5	1	1	0	0	0	0	0	1
6	0	0	1	1	0	1	1	1

For $i = 1, ..., n$, let $Y_i \in \{0, 1\}$ indicate army-worm damage observed in cell i. The data are displayed above. Suppose $Y_i \sim \text{Bernoulli}(\mu_i)$ independently in each cell, with

$$\mu_i = \frac{\exp(z_i)}{1 + \exp(z_i)}.$$

where $z_i \in \Re$ is an unobserved real latent variable associated with cell i. Let $z = (z_1, ..., z_n)$. The z-values are modelled by an auto-regression with an $n \times n$ weight matrix W

$$z = (I - W)^{-1}\epsilon, \quad \epsilon \sim N(0, \sigma^2 I_n)$$

It is of interest to estimate a map of the log-odds for damage, and to test for an effect due to ploughing. Assume a prior for σ is available.

(a) [9 marks] Consider eliciting a prior for z by specifying W.

(i) Suppose that for $i = 1, ..., n$, $W_{ii} = 0$. Let $\pi(z_i|z_{-i}, \sigma)$ denote the conditional density of z_i given the other z-values and σ. Show that

$$p(z_i|z_{-i}, \sigma) = N(z_i; \textstyle\sum_{j \neq i} W_{i,j} z_j, \sigma^2).$$

(ii) Scientist A believes a small number of army worm midges blew in after planting, so that the first arrivals were scatted at random over the area and these then expanded through the ground to form local clusters. Using the auto-regression, elicit a prior $\pi_A(z|\sigma, u)$ for z with a single scalar parameter u.

(iii) Scientist B believes the army worms blew in before planting and formed small clusters a foot or so across which were subsequently smeared out along the columns by ploughing and seeding. Using the auto-regression, elicit a prior $\pi_B(z|\sigma, v)$ for z with a single scalar parameter v.

(iv) Suppose priors for σ, u and v have been elicited. Say how you would check that these priors and the priors for z reflect available knowledge. Indicate any further information you would need from Scientists A and B.

(b) [11 marks] Assume independent priors $f(\sigma)$, $g(u)$ and $h(v)$ are given.

(i) Let $m = A, B$ indicate the model and let $u = \theta$ if $m = A$ and $v = \theta$ if $m = B$. Elicit a prior for the model index m and write down the posterior distribution $\pi(z, m, \sigma, \theta|y)$ for the model and parameters in as much detail as you can.

(ii) Give the z and m updates of an MCMC algorithm targeting $\pi(z, m|y)$. You should give the proposals and acceptance probabilities in as much detail as you can.

(iii) Outline briefly how to tune your MCMC for efficiency and check for convergence.

(iv) The MCMC output is $m^{(t)}, z^{(t)}$, $t = 1, ..., T$. Explain how to do model selection and how to estimate $E(Z|Y = y)$.

2. Data on insects caught over several nights in a mesh net are recorded. Let $N_{i,j}$ give the number of insects of species $i = 1, ..., n$ caught on night $j = 1, ..., J$. Let $\lambda_{i,j}$ give the arrival rate per night for species i and let β_i and γ_j be random effects for species and night. In a log-linear model $log(\lambda_{i,j}) = \alpha + \beta_i + \gamma_j$ and $N_{i,j} \sim \text{Poisson}(\lambda_{i,j})$ independently. Let $N = (N_{i,j})_{i=1,...,n}^{j=1,...,J}$.

It is thought likely that many of the β-values are equal for groups of insects with similar abundance and behavior. Let $\beta^* = (\beta_1^*, ..., \beta_K^*)$ denote the K distinct β-values (with K unknown). Let $\gamma = (\gamma_1, ...\gamma_J)$. The prior distribution for the J random effects is $\gamma \sim N(0, \sigma_g^2 I_J)$. Assume priors $\pi(\sigma, \sigma_g)$ are given.

(a) [10 marks] In this part we consider a Dirichlet process model for the grouping with $S = (S_1, ..., S_K)$ a K-set partition of $\{1, ..., n\}$ having a Chinese Restaurant Process prior distribution $P_\alpha(S)$ with given parameter α determining the distribution of K. If species i and j are in the same set, so that $i, j \in S_k$ for some k in $1, ..., K$, then $\beta_i = \beta_j = \beta_k^*$.

(i) Write down, in as much detail as you can, the posterior $\pi(S, \beta^*, \gamma, \sigma, \sigma_g|N)$ with base distribution $\beta^* \sim N(0, \sigma^2 I_K)$ and concentration parameter α.

(ii) Give the S update for a Gibbs sampler targeting $\pi(S, \beta^*, \gamma, \sigma, \sigma_g|N)$.

(b) [10 marks] In a new model the unknown number of clusters has a Poisson distribution $K \sim \text{Poisson}(M)$ with M fixed. Let $w = (w_1, ..., w_K)$ be a vector of positive weights, normalised so that $\sum_k w_k = 1$ and having prior $w \sim \text{Dirichlet}(\alpha/M, ..., \alpha/M)$. Consider the same data, but suppose the observation model is now a mixture,

$$N_{i,j} \sim \sum_{k=1}^{K} w_k \text{Poisson}(\lambda_{k,j}^*), \text{ independently for } i = 1, ..., n \text{ and } j = 1, ..., J,$$

with $\lambda_{k,j}^* = \exp(\alpha + \beta_k^* + \gamma_j)$. The priors for $\beta^*|K$, γ, σ and σ_g are unchanged.

(i) Write down, in as much detail as you can, the posterior $\pi(w, \beta^*, \gamma, \sigma, \sigma_g|N)$ in terms of the likelihood and parameter priors (note that $K = \dim(\beta^*)$ is known given β^* so we make no separate reference to this parameter).

(ii) In a reversible jump MCMC algorithm targeting $\pi(w, \beta^*, \gamma, \sigma, \sigma_g|N)$, the proposals to add or drop an element of $\beta^* = (\beta_1^*, ..., \beta_K^*)$ are chosen with probabilities $\rho_{K,K+1}$ and $\rho_{K,K-1}$ respectively. Give proposals for adding and removing elements of β^* and the acceptance probability for the move to add a dimension. Distributions must be well defined, but there is no need to simplify in acceptance probabilities.